

CBSE Practice Worksheet

Class 6 | Mathematics | Difficulty: Hard | Set B

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Find: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$
(A) 2
(B) 8
(C) 15
(D) $\sqrt{15}$
2. Solve: $3(x-2)/4 = (2x+1)/3$
(A) $x = 14$
(B) $x = 22$
(C) $x = 26$
(D) $x = 30$
3. Solve: $|2x - 5| = 9$
(A) $x = 7$ or -2
(B) $x = 7$ or 2
(C) $x = -7$ or 2
(D) $x = -7$ or -2
4. Simplify: $(a^2 - b^2)/(a + b)$
(A) $a - b$
(B) $a + b$
(C) $a^2 - b$
(D) $a - b^2$
5. Factorize: $x^2 - 5x - 14$
(A) $(x-7)(x+2)$
(B) $(x+7)(x-2)$
(C) $(x-7)(x-2)$
(D) $(x+7)(x+2)$
6. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
(A) $x - 2$
(B) $x + 2$
(C) $(x-2)^2$
(D) $x^2 - 4$
7. What is the cube root of -125 ?
(A) 5
(B) -5
(C) ± 5
(D) undefined
8. If $a:b = 3:4$ and $b:c = 5:6$, find $a:c$
(A) $3:6$
(B) $5:8$
(C) $15:24$

(D) 1:2

9. If $2^x = 3^y = 6^z$, which is true?

(A) $1/z = 1/x + 1/y$

(B) $z = x + y$

(C) $z = xy$

(D) $z = x - y$

10. If $x + 1/x = 5$, find $x^2 + 1/x^2$

(A) 21

(B) 23

(C) 25

(D) 27

Section B: Fill in the Blanks (1 mark each)

11. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 = \underline{\hspace{2cm}}$

12. Rationalize: $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/\underline{\hspace{2cm}}$

13. If $5^x = 125$, then $x = \underline{\hspace{2cm}}$

14. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - \underline{\hspace{2cm}}$

15. $\sqrt[3]{-64} = \underline{\hspace{2cm}}$

16. $\text{HCF} \times \text{LCM}$ of 12 and 18 = $\underline{\hspace{2cm}}$

17. If $x^2 - 6x + 8 = 0$, then $x = \underline{\hspace{2cm}}$ or $\underline{\hspace{2cm}}$

18. If $\log x = 2$, then $x = \underline{\hspace{2cm}}$

Section C: Short Answer Questions (2 marks each)

19. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

20. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

21. Factorize: $8x^3 - 27$

22. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

23. Find the value of x : $3^{(2x-1)} = 27^{(x-2)}$

24. Solve: $\log_{10}(x - 1) + \log_{10}(x + 1) = 3$

25. If $x = 2 + \sqrt{3}$, find $x + 1/x$

--- End of Worksheet ---

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